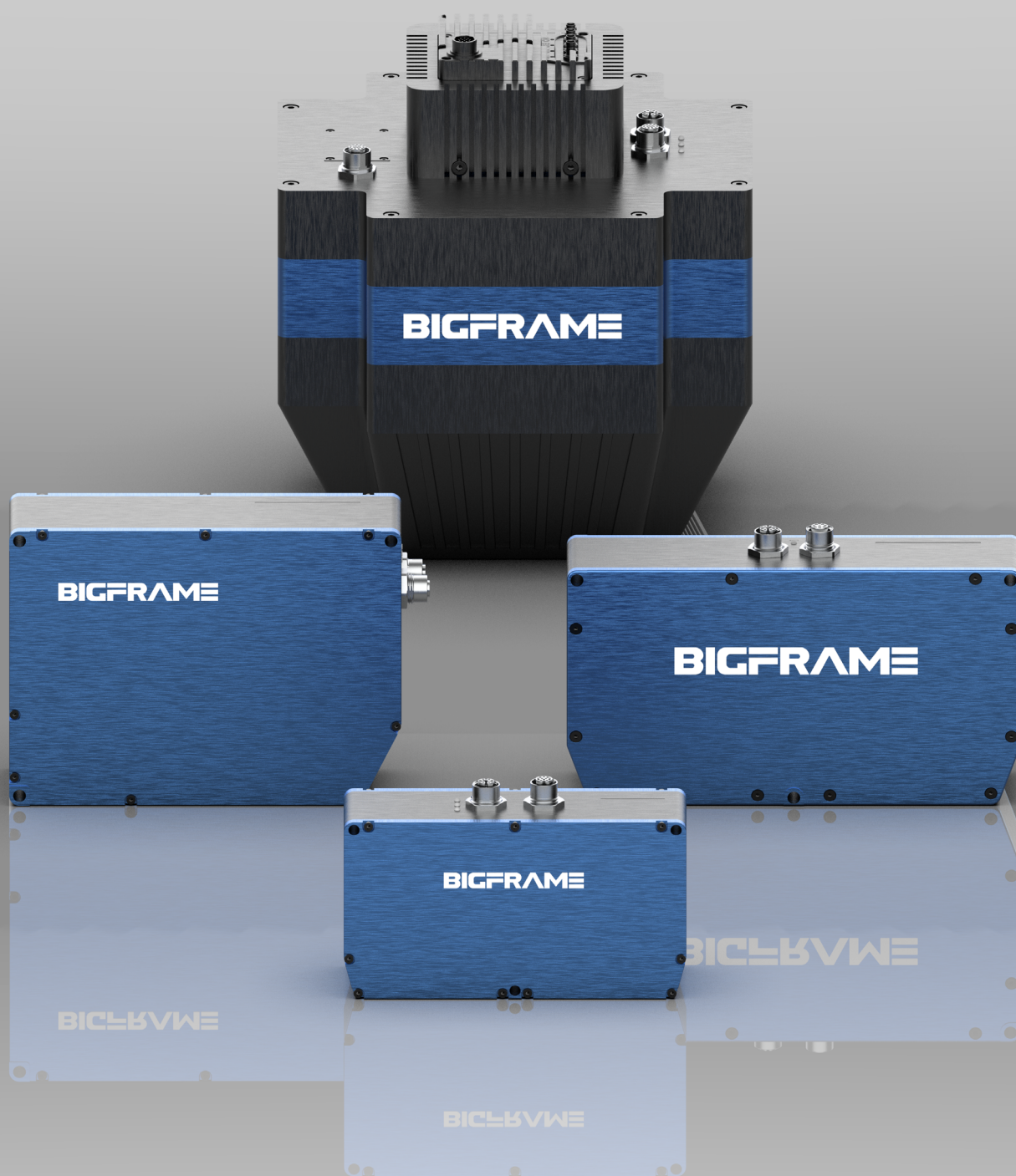


BIGFRAME
大帧科技

IMALLIGENT
珩图科技

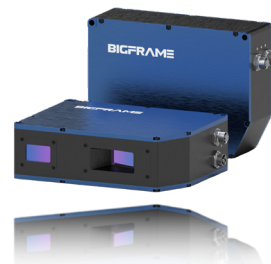


PASSION FOR 3D

High-accuracy Velociraptor | VRH9 Series

Area-scanning Structured Light 3D Camera

- Featuring 9.07 megapixels and high-precision imaging, this system delivers a frame rate of up to 6.6 frames per second. It supports NVIDIA CUDA, offers a flexible field of view, and achieves a Z-axis regional repeatability accuracy of up to 0.15 μm , making it ideal for high-precision industrial inspection applications.



PRODUCTS SPEC

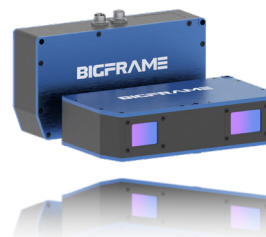
Model	VRH9-020B	VRH9-040B	VRH9-080B	VRH9-120B	VRH9-160B	VRH9-200B	VRH9-280B
Resolution	9.07M 3D Point Cloud						
Scan Range (Z-axis)	112 ~ 114.5 mm	243 ~ 257 mm	260 ~ 280 mm	260 ~ 290 mm	340 ~ 380 mm	330 ~ 390 mm	450 ~ 550 mm
Near-field (WD, mm)	24.1 x 12.4 mm (112)	47 x 24 mm (243)	79 x 40 mm (260)	120 x 61.5 mm (260)	151 x 77 mm (340)	190 x 98 mm (330)	251 x 129 mm (450)
Mid-field (WD, mm)	24.3 x 12.5 mm (113)	48.2 x 24.7 mm (250)	81.5 x 41.5 mm (270)	126 x 64.5 mm (275)	159 x 81.5 mm (360)	205 x 105 mm (360)	277 x 142 mm (500)
Far-field (WD, mm)	24.6 x 12.6 mm (114.5)	49.5 x 25.5 mm (257)	84 x 43 mm (280)	132 x 67.5 mm (290)	167 x 85 mm (380)	220 x 112 mm (390)	302 x 155 mm (550)
Resolution, XY-axis	5.8 μm	11 μm	19 μm	30 μm	38 μm	49 μm	66 μm
Z-axis Regional Repeatability ⁽¹⁾⁽²⁾	0.15 μm	0.23 μm	0.36 μm	0.5 μm	0.95 μm	1.3 μm	1.8 μm
Optimal Scanning Volume	24.3 x 12.5 x 2.5 mm	48.2 x 24.7 x 14 mm	81.5 x 41.5 x 20 mm	126 x 64.5 x 30 mm	159 x 81.5 x 40 mm	205 x 105 x 60 mm	277 x 142 x 100 mm
Scan Rate	1.1 fps @ (4200 x 2160), 6.6 fps @ (640 x 480)						
Power	$\geq 75\text{ W}$						
Input Voltage	DC 24 V						
Wavelength	Blue Light						
Data Interface	Gigabit Ethernet						
Size	190 x 140 x 60 mm						
Weight	1800 g						

(1) Z-axis Regional Repeatability: The maximum value of the 2-sigma Z-axis repeatability obtained from measurements across 9 evenly divided regions within the Field of View (FOV), when imaging a target ceramic plate at near-field, mid-field, and far-field working distances, respectively.
(2) The 1-sigma standard deviation of 100 consecutive measurements of the difference between the average Z-value of Region A and the average Z-value of Region B. Both Region A and Region B are adjacent, and each has an area less than 1% of the total Field of View (FOV).

Velociraptor | VR Series

Area-scanning Structured Light 3D Camera

- Featuring NVIDIA CUDA cores and DLP projection, this system delivers a scanning speed of up to 4 frames per second, with a field of view ranging from 215 mm to 2180 mm and a Z-axis regional repeatability accuracy of up to 2 μm , making it perfect for high-definition 3D imaging across diverse scenarios.



PRODUCTS SPEC

Model	VR-400B	VR-600B	VR-1300B	VR-2300B
Resolution	3.2M 3D Point Cloud			
Scan Range (Z-axis)	350 ~ 650 mm	500 ~ 900 mm	900 ~ 2200 mm	1900 ~ 3800 mm
Near-field (WD, mm)	215 x 160 mm (350)	300 x 225 mm (500)	530 x 395 mm (900)	1100 x 820 mm (1900)
Mid-field (WD, mm)	300 x 220 mm (500)	415 x 305 mm (700)	865 x 645 mm (1500)	1500 x 1115 mm (2600)
Far-field (WD, mm)	380 x 285 mm (650)	525 x 385 mm (900)	1250 x 940 mm (2200)	2180 x 1625 mm (3800)
Resolution, XY-axis	0.15 mm	0.20 mm	0.42 mm	0.73 mm
Z-axis Regional Repeatability ⁽¹⁾⁽²⁾	2 μm	6 μm	34 μm	100 μm
Optimal Scanning Volume	300 x 220 x 150 mm	415 x 305 x 200 mm	865 x 645 x 700 mm	1500 x 1115 x 1200 mm
Scan Rate	1.1 fps @ (2048 x 1536), 4 fps @ (640 x 480)			
Power	$\geq 75\text{ W}$			
Input Voltage	DC 24 V			
Wavelength	Blue Light			
Data Interface	Gigabit Ethernet			
Size	220 x 120 x 56 mm			
Weight	1600 g			

(1) Z-axis Regional Repeatability: The maximum value of the 2-sigma Z-axis repeatability obtained from measurements across 9 evenly divided regions within the Field of View (FOV), when imaging a target ceramic plate at near-field, mid-field, and far-field working distances, respectively.
(2) The 1-sigma standard deviation of 100 consecutive measurements of the difference between the average Z-value of Region A and the average Z-value of Region B. Both Region A and Region B are adjacent, and each has an area less than 1% of the total Field of View (FOV).

Area-scanning Structured Light 3D Camera

Tyrannosaurus Rex | TRS Series

Area-scanning Structured Light 3D Camera

- Powered by NVIDIA CUDA for accelerated imaging calculations, this system offers four-projection imaging with no dead spots, a 9.8- 25.5 mm field of view, and a Z-axis repeatability accuracy of 0.06 μm , making it perfect for precise detection of extremely small targets.



PRODUCTS SPEC

Model	TRS-010B	TRS-025B
Resolution	26M 3D Point Cloud	
Scan Range (Z-axis)	1 mm	5 mm
Near-field (WD, mm)	9.8 x 9.8 mm (62)	25.5 x 25.5 mm (101.5)
Mid-field (WD, mm)	9.8 x 9.8 mm (62.5)	25.5 x 25.5 mm (104)
Far-field (WD, mm)	9.8 x 9.8 mm (63)	25.5 x 25.5 mm (106.5)
Resolution, XY-axis	1.9 μm	5 μm
Z-axis Regional Repeatability ⁽¹⁾⁽²⁾	0.06 μm	0.1 μm
Optimal Scanning Volume	9.8 x 9.8 x 1 mm	25.5 x 25.5 x 5 mm
Scan Rate	≤ 3 fps	
Power	≥ 75 W	
Input Voltage	DC 24 V	
Wavelength	Blue Light	
Data Interface	CXP	
Size	230 x 230 x 250 mm	240 x 240 x 275 mm
Weight	8850 g	9740 g

(1) Z-axis Regional Repeatability: The maximum value of the 2-sigma Z-axis repeatability obtained from measurements across 9 evenly divided regions within the Field of View (FOV), when imaging a target ceramic plate at near-field, mid-field, and far-field working distances, respectively.
(2) The 1-sigma standard deviation of 100 consecutive measurements of the difference between the average Z-value of Region A and the average Z-value of Region B. Both Region A and Region B are adjacent, and each has an area less than 1% of the total Field of View (FOV).

Color Velociraptor Stereo | VRD Series

Area-scanning Structured Light 3D Camera

- The device uses DLP structured light technology with 5MP resolution and binocular 3D imaging to minimize occlusion, offering RGB image output. Its compact design is ideal for 3D guidance of collaborative robots and surface inspection.



PRODUCTS SPEC

Model	VRD-600B	VRD-1200B
Resolution	5M 3D Point Cloud	
Scan Range (Z-axis)	400 ~ 800 mm	900 ~ 1600 mm
Near-field (WD, mm)	365 x 248 mm (400)	887 x 551 mm (900)
Mid-field (WD, mm)	580 x 369 mm (600)	1166 x 731 mm (1200)
Far-field (WD, mm)	795 x 490 mm (800)	1538 x 974 mm (1600)
Resolution, XY-axis	0.24 mm	0.48 mm
Z-axis Regional Repeatability ⁽¹⁾⁽²⁾	6 μm	24 μm
Optimal Scanning Volume	580 x 369 x 300 mm	1166 x 731 x 400 mm
Scan Rate	≤ 1.2 fps	
Power	≥ 75 W	
Input Voltage	DC 24 V	
Wavelength	Blue Light	
Data Interface	Gigabit Ethernet	
Size	166 x 90 x 54 mm	
Weight	870 g	

(1) Z-axis Regional Repeatability: The maximum value of the 2-sigma Z-axis repeatability obtained from measurements across 9 evenly divided regions within the Field of View (FOV), when imaging a target ceramic plate at near-field, mid-field, and far-field working distances, respectively.
(2) The 1-sigma standard deviation of 100 consecutive measurements of the difference between the average Z-value of Region A and the average Z-value of Region B. Both Region A and Region B are adjacent, and each has an area less than 1% of the total Field of View (FOV).



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